

Inverse of 3x3 matrices

Basic skills

Q1. a) $A = \begin{pmatrix} 4 & -2 \\ 1 & 7 \end{pmatrix}$ $|A| = 28 - (-2) = 30$
 $\text{Adj}(A) = \begin{pmatrix} 7 & 2 \\ -1 & 4 \end{pmatrix}$

so $A^{-1} = \frac{1}{|A|} \text{Adj}(A) = \frac{1}{30} \begin{pmatrix} 7 & 2 \\ -1 & 4 \end{pmatrix}$

b) $B = \begin{pmatrix} 6 & 8 \\ 3 & 4 \end{pmatrix}$ $|B| = 24 - 24 = 0$
so no inverse exists

Q2. A non-zero determinant i.e. $|M| \neq 0$

Q3. $M = \begin{pmatrix} 6 & 5 & -3 \\ 3 & 0 & 2 \\ -2 & 1 & 10 \end{pmatrix}$

a) $M_{1,3} = \begin{pmatrix} 3 & 0 \\ -2 & 1 \end{pmatrix}$ $M_{2,2} = \begin{pmatrix} 6 & -3 \\ -2 & 10 \end{pmatrix}$

b) $C_{13} = |M_{1,3}| = 3 - 0 = 3$ $C_{22} = |M_{2,2}| = 60 + 6 = 66$

c) $|M| = 6 \begin{vmatrix} 0 & 2 \\ 1 & 10 \end{vmatrix} - 5 \begin{vmatrix} 3 & 2 \\ -2 & 10 \end{vmatrix} + (-3) \begin{vmatrix} 3 & 0 \\ -2 & 1 \end{vmatrix}$

$$\Rightarrow |M| = 6(0-2) - 5(30+4) - 3(3-0) \\ = -12 - 170 - 9 = -191$$

d) so M is invertible as $|M| \neq 0$

Competency skills

Q 4. $M = \begin{pmatrix} 3 & 0 & 1 \\ 2 & 0 & -2 \\ 0 & 5 & 1 \end{pmatrix}$

a) $|M| = 3 \begin{vmatrix} 0 & -2 \\ 5 & 1 \end{vmatrix} - 0 + 1 \begin{vmatrix} 2 & 0 \\ 0 & 5 \end{vmatrix} \\ = 3(0-10) + 10 = 40$

b)

$$|M_{1,1}| = \begin{vmatrix} 0 & -2 \\ 5 & 1 \end{vmatrix} = 10 \quad |M_{2,3}| = \begin{vmatrix} 3 & 0 \\ 0 & 5 \end{vmatrix} = 15$$

$$|M_{1,2}| = \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} = 2 \quad |M_{3,1}| = \begin{vmatrix} 0 & 1 \\ 0 & -2 \end{vmatrix} = 0$$

$$|M_{1,3}| = \begin{vmatrix} 2 & 0 \\ 0 & 5 \end{vmatrix} = 10 \quad |M_{3,2}| = \begin{vmatrix} 3 & 1 \\ 2 & -2 \end{vmatrix} = -6-2 = -8$$

$$|M_{2,1}| = \begin{vmatrix} 0 & 1 \\ 5 & 1 \end{vmatrix} = -5 \quad |M_{3,3}| = \begin{vmatrix} 3 & 0 \\ 2 & 0 \end{vmatrix} = 0$$

$$|M_{2,2}| = \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} = 3$$

✓ $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$

so $C = \begin{pmatrix} 10 & -2 & 10 \\ 5 & 3 & -15 \\ 0 & 8 & 0 \end{pmatrix}$

↓ transpose

swap row, col

c) $\text{Adj}(M) = C^T = \begin{pmatrix} 10 & 5 & 0 \\ -2 & 3 & 8 \\ 10 & -15 & 0 \end{pmatrix}$

so $M^{-1} = \frac{1}{40} \begin{pmatrix} 10 & 5 & 0 \\ -2 & 3 & 8 \\ 10 & -15 & 0 \end{pmatrix}$

d) $M M^{-1} = \frac{1}{40} \begin{pmatrix} 3 & 0 & 1 \\ 2 & 0 & -2 \\ 0 & 5 & 1 \end{pmatrix} \begin{pmatrix} 10 & 5 & 0 \\ -2 & 3 & 8 \\ 10 & -15 & 0 \end{pmatrix}$

$$= \frac{1}{40} \begin{pmatrix} 40 & 0 & 0 \\ 0 & 40 & 0 \\ 0 & 0 & 40 \end{pmatrix} = I_3$$

so yes the inverse is correct

Q5.

a) $A = \begin{pmatrix} 6 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 5 & -6 \end{pmatrix}$

$$|A| = 6 \begin{vmatrix} -1 & 0 \\ 5 & -6 \end{vmatrix} = 6 \times (6 - 0) = 36$$

Then

$$|A_{1,1}| = \begin{vmatrix} -1 & 0 \\ 5 & -6 \end{vmatrix} = 6 \quad |A_{2,3}| = \begin{vmatrix} 6 & 0 \\ 0 & 5 \end{vmatrix} = 30$$

$$|A_{1,2}| = \begin{vmatrix} 0 & 0 \\ 0 & -6 \end{vmatrix} = 0 \quad |A_{3,1}| = \begin{vmatrix} 0 & 0 \\ -1 & 0 \end{vmatrix} = 0$$

$$|A_{1,3}| = \begin{vmatrix} 0 & -1 \\ 0 & 5 \end{vmatrix} = 0$$

$$|A_{3,2}| = \begin{vmatrix} 6 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$|A_{2,1}| = \begin{vmatrix} 0 & 0 \\ 5 & -6 \end{vmatrix} = 0$$

$$|A_{3,3}| = \begin{vmatrix} 6 & 0 \\ 0 & -1 \end{vmatrix} = -6$$

$$|A_{2,2}| = \begin{vmatrix} 6 & 0 \\ 0 & -6 \end{vmatrix} = -36$$

so applying alternating sign

$$C = \begin{pmatrix} 6 & 0 & 0 \\ 0 & -36 & -30 \\ 0 & 0 & -6 \end{pmatrix} \quad \text{so } C^T = \begin{pmatrix} 6 & 0 & 0 \\ 0 & -36 & 0 \\ 0 & -30 & -6 \end{pmatrix}$$

$$\text{so } A^{-1} = \frac{1}{36} \begin{pmatrix} 6 & 0 & 0 \\ 0 & -36 & 0 \\ 0 & -30 & -6 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & -5 & -1 \end{pmatrix}$$

$$b) B = \begin{pmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{pmatrix}$$

$$\begin{aligned} |B| &= 1 \begin{vmatrix} -1 & -6 \\ 0 & 6 \end{vmatrix} - 2 \begin{vmatrix} 2 & -6 \\ -2 & 6 \end{vmatrix} - 3 \begin{vmatrix} 2 & -1 \\ -2 & 0 \end{vmatrix} \\ &= 1(-6-0) - 2(12-12) - 3(-2) \\ &= -6 + 6 = 0 \end{aligned}$$

So not invertible

$$c) C = \begin{pmatrix} 1 & 5 & 3 \\ 2 & 4 & -2 \\ 4 & 2 & 1 \end{pmatrix}$$

$$\begin{aligned} |C| &= 1 \begin{vmatrix} 4 & -2 \\ 2 & 1 \end{vmatrix} - 5 \begin{vmatrix} 2 & -2 \\ 4 & 1 \end{vmatrix} + 3 \begin{vmatrix} 2 & 4 \\ 4 & 2 \end{vmatrix} \\ &= (4+4) - 5(2+8) + 3(4-16) \\ &= 8 - 50 - 36 = -78 \end{aligned}$$

Then minors

$$|M_{1,1}| = 8 \quad |M_{1,2}| = 10 \quad |M_{1,3}| = -12$$

$$|M_{2,1}| = \begin{vmatrix} 5 & 3 \\ 2 & 1 \end{vmatrix} = -1 \quad |M_{3,1}| = \begin{vmatrix} 5 & 3 \\ 4 & -2 \end{vmatrix} = -22$$

$$|M_{2,2}| = \begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix} = -11 \quad |M_{3,2}| = \begin{vmatrix} 1 & 3 \\ 2 & -2 \end{vmatrix} = -8$$

$$|M_{2,3}| = \begin{vmatrix} 1 & 5 \\ 4 & 2 \end{vmatrix} = -18 \quad |M_{3,3}| = \begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} = -6$$

so cofactors is

$$\begin{pmatrix} 8 & -10 & -12 \\ 1 & -11 & 18 \\ -22 & 8 & -6 \end{pmatrix}$$

$$\text{so } \text{Adj}(C) = \begin{pmatrix} 8 & 1 & -22 \\ -10 & -11 & 8 \\ -12 & 18 & -6 \end{pmatrix}$$

$$\begin{aligned} \text{and } M^{-1} &= -\frac{1}{78} \begin{pmatrix} 8 & 1 & -22 \\ -10 & -11 & 8 \\ -12 & 18 & -6 \end{pmatrix} \\ &= \frac{1}{78} \begin{pmatrix} -8 & -1 & 22 \\ 10 & 11 & -8 \\ 12 & -18 & 6 \end{pmatrix} \end{aligned}$$

Q6. $A = \begin{pmatrix} 1 & -2 & k \\ 2 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $B = \begin{pmatrix} -5 & -2+2k & -4-k \\ 8 & -1-3k & -2+2k \\ 1 & -8 & 5 \end{pmatrix}$

a) $|A| = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 2 \\ 3 & -1 \end{vmatrix} + k \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix}$

$= -5 + 2 \times -8 + k \times 1 = k - 21$

If A^{-1} possible $|A| \neq 0$ so $k - 21 \neq 0$

$\Rightarrow k \neq 21$ ($k > 21$ or $k < 21$)

b) $AB = \begin{pmatrix} 1 & -2 & k \\ 2 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix} \begin{pmatrix} -5 & -2+2k & -4-k \\ 8 & -1-3k & -2+2k \\ 1 & -8 & 5 \end{pmatrix}$

$= \begin{pmatrix} k-21 & 0 & 0 \\ 0 & k-21 & 0 \\ 0 & 0 & k-21 \end{pmatrix}$

$= (k-21) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

so

$\frac{1}{k-21} AB = I_3$

So

$$\frac{1}{k-21} B = A^{-1}$$

$$A^{-1} = \frac{1}{k-21} \begin{pmatrix} -5 & -2+2k & -4-k \\ 8 & -1-3k & -2+2k \\ 1 & -8 & 5 \end{pmatrix}$$

$$c) A\underline{v} = \begin{pmatrix} 1 \\ 12 \\ 3 \end{pmatrix} \Rightarrow \underline{v} = A^{-1} \begin{pmatrix} 1 \\ 12 \\ 3 \end{pmatrix}$$

$$\text{So } \underline{v} = \frac{1}{k-21} \begin{pmatrix} -5 & -2+2k & -4-k \\ 8 & -1-3k & -2+2k \\ 1 & -8 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 12 \\ 3 \end{pmatrix}$$

$$= \frac{1}{k-21} \begin{pmatrix} 21k-42 \\ -30k-10 \\ -80 \end{pmatrix}$$

$$d) \text{ solved } \begin{pmatrix} 1 & -2 & k \\ 2 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 12 \\ 3 \end{pmatrix}$$

$$\Rightarrow x - 2y + kz = 1$$

$$2x + y + 2z = 12$$

$$3x + 2y - z = 3$$